

ARAV GUPTA

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LinkedIn | GitHub | LeetCode | HackerRank

PROFESSIONAL SUMMARY

Scalable Systems Architect, **Android & iOS Developer**, and Full Stack Engineer with expertise in **Distributed Ledger Technology (DLT)** and **Generative AI** orchestration. Advanced proficiency in **Java 21 (Basic to Expert)**, **Rust**, and **Swift**. Expert in designing high-concurrency microservices and **RAG pipelines** using **Next.js 15** and **Node.js 22**. Proven lead in orchestrating **GKE** and **AWS** infrastructure to deliver enterprise-grade reliability, security, and 99.9% uptime for complex, multi-stakeholder applications.

TECHNICAL SKILLS

Languages: Java 21 (Advanced), Rust, Swift, Kotlin, TypeScript, JavaScript (ES6+), Python 3.12, C++, Dart, Solidity

Mobile & Frontend: Android Studio, iOS Development, Flutter 3.x, Next.js 15, React 19, Tailwind CSS 4, HTML5, CSS3

Backend & Cloud: Node.js 22, Express.js, Spring Boot 3.x, AWS (EC2, S3, Lambda, IAM), Google Cloud (GKE), Vercel

Databases & AI: MongoDB, PostgreSQL 16, Pinecone, Milvus (Vector DB), Gemini 1.5 Pro, Claude 3.5 Sonnet, LangGraph (RAG)

Engineering & Security: Distributed Systems, Microservices, RESTful APIs, Docker, CI/CD, Penetration Testing, JWT/OAuth 2.0

PROFESSIONAL EXPERIENCE

Freelance Software Engineer

Remote | Jan 2024 – Present

Full Stack Web & Mobile Development

- Architected and delivered end-to-end web and mobile solutions using **Next.js 15**, **Flutter**, and **Supabase**, delivering production-grade applications that enhanced client operational efficiency by 40%.
- Translated complex business requirements into technical architectures, implementing secure payment gateways (Stripe/Razorpay) and authenticated user sessions via **NextAuth.js**.
- Automated operational workflows by integrating **Gemini 1.5 Pro** and **Claude 3.5 APIs**, reducing manual effort for client-side business processes by 50% through autonomous AI agents.
- Provided strategic consultation on **Blockchain and System Design**, ensuring scalable, high-performance infrastructures for startups and enterprise-level clients.

TECHNICAL PROJECTS

Rakt-Connect | Blockchain Supply Chain & Emergency Broadcast

Lead Engineer & Architect

Hyperledger Fabric, Next.js 15, Node.js 22, GKE, MongoDB, Google Cloud

- Architected a permissioned blockchain network using **Hyperledger Fabric**, ensuring 100% data immutability and end-to-end traceability for healthcare supply chains.
- Optimized system throughput to support **1,000+ concurrent requests** by implementing efficient MongoDB indexing and container orchestration on **GKE**.
- Developed smart contracts (Chaincode) to automate donor verification, reducing administrative latency by 90% and eliminating duplicate record vulnerabilities.

AI Trading Bot | Quantitative Analysis & Data Engineering

Python, TensorFlow, Groq API, Gemini 1.5 Pro, Pandas

- Built an **LSTM-based time-series model** achieving 80% directional accuracy on 10,000+ real-time financial data points.
- Utilized **Groq API** for ultra-fast LPU inference and **Gemini 1.5 Pro** for sentiment analysis, reducing signal detection latency to **<30ms**.

AI Slop Detector | GenAI & NLP Orchestration

Tavily AI, LangChain, LangGraph, RAG, Vertex AI

- Designed an autonomous agent using **LangChain** and **LangGraph** to classify AI-generated content with 85%+ accuracy via **RAG pipelines** and Pinecone vector indexing.
- Integrated **Tavily AI** for real-time web-search verification, improving model robustness against AI hallucinations during fact-checking simulations.

AI SaaS Project Management Platform

Next.js 15, Node.js 22, MongoDB, Clerk, OpenAI API

- Engineered a multi-tenant platform supporting real-time state synchronization for 100+ simulated concurrent users.
- Integrated AI-driven task prioritization and automated summaries, improving user productivity metrics by 40%.

EDUCATION

Visvesvaraya Technological University (VTU)

Bengaluru, India | 2027

B.E. in Computer Science (Sambhram Institute of Technology)

Swaraj India Public School

ISC Class XII (72.75%) | 2022

Swaraj India Public School

ICSE Class X (87.8%) | 2020

CERTIFICATIONS & TRAINING

- Placement Training (60 Hours):** Advanced technical/soft skills curriculum by Genesis Training.
- Vtricks Technologies:** Java, SQL, NoSQL Database Design, and Data Analytics (Python/PowerBI).
- COMED KARES:** Innovation & Design Thinking Program (6-month cohort).